Math 120 Spring 2020 - Quiz 12

Discussion Time: **Name:**

Do not use calculators.

Honor Pledge: I pledge on my honor that I have (and will) not give or receive any unauthorized assistance on this examination. (Copy and sign.)

1. (2 pts) Find the equation of the line through the points $(1, 4)$ and $(3, 8)$. Your answer should be in the form $y = mx + b$.

2. (4 pts) Suppose that the amount of time that a math student spends on the Taco Bell during class break has probability density function
$$f(t) = \frac{2}{225}(15 - t) \quad 0 \leq t \leq 15.$$
Find the probability that a random student spends between 5 and 10 minutes.

3. (4 pts) Let $f(x, y) = x^2 e^{3xy}$. Compute
$$\frac{\partial f(x,y)}{\partial x} \quad \text{and} \quad \frac{\partial \mathrm{f(x,y)}}{\partial \mathrm{y}}.$$